

NumPy Question Paper

Easy (5)

1. Create a NumPy array containing the values [10, 20, 30, 40, 50]. Print its shape and data type.
2. Write a program to create a 3×4 array of zeros and a 2×3 array of ones.
3. Using np.arange() and np.linspace(), generate numbers from 1 to 10 and 5 evenly spaced values between 0 and 100.
4. Given arr = np.array([5,10,15,20,25,30]), print the first three elements, last two elements, and every second element.
5. Given marks = np.array([[80,90,70],[75,85,95],[60,88,92]]), print the second row and third column.

Medium (3)

6. Create a 1D array with values 1 to 12. Reshape it into (3,4), (2,6), and (4,3).
7. Given scores = np.array([45,82,91,38,76,95,60]), use boolean indexing to print only scores greater than or equal to 75 and count them.
8. Given two arrays a=[10,20,30,40] and b=[2,4,6,8], perform element-wise addition, subtraction, multiplication, and division.

Hard (3)

9. Given a 4×3 marks matrix and bonus = np.array([5,3,2]), use broadcasting to add subject-wise bonus marks to every student.
10. For a 2D NumPy array of student marks, calculate row-wise totals, row-wise averages, column-wise averages, overall maximum, minimum, and standard deviation.
11. Create a 3×4 matrix and a (4,1) weight matrix. Use matrix multiplication (@) to calculate the weighted score of each row.